

SHOPSY E-COMMERCE DATABASE MANAGEMENT SYSTEM

PAYMENT MANAGEMENT SYSTEM

1. Database Design :

The Payment entity is designed to store and manage payment transaction details for customer orders. It records information such as payment ID, order ID, payment method, payment status, payment date, and payment amount. The Payment_ID uniquely identifies each payment transaction and acts as the Primary Key. The Order_ID connects the Payment table with the Orders table and acts as a Foreign Key. The Payment table helps to manage successful and failed transactions and provides useful information for analyzing payment methods and generating payment transaction reports.

1.1 Payment Table :

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	-
Payment_Date	-
Amount	-

2. Create Payment Table :

```
SQL> CREATE TABLE Payment (  
  2     Payment_ID NUMBER PRIMARY KEY,  
  3     Order_ID NUMBER,  
  4     Payment_Method VARCHAR2(30),  
  5     Payment_Status VARCHAR2(20),  
  6     Payment_Date DATE,  
  7     Amount NUMBER(10,2),  
  8     FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID)  
  9 );
```

Table created.

3. Payment Records :

```
SQL> INSERT INTO Payment VALUES  
  2 (1, 1001, 'Credit Card', 'Successful', DATE '2026-09-01', 2098.00);
```

1 row created.

```
SQL> INSERT INTO Payment VALUES  
  2 (2, 1002, 'UPI', 'Successful', DATE '2026-09-03', 1499.00);
```

1 row created.

```
SQL> INSERT INTO Payment VALUES  
  2 (3, 1003, 'Debit Card', 'Successful', DATE '2026-09-05', 2598.00);
```

1 row created.

```
SQL> INSERT INTO Payment VALUES  
  2 (4, 1004, 'Cash on Delivery', 'Successful', DATE '2026-09-07', 1999.00);
```

1 row created.

```
SQL> INSERT INTO Payment VALUES  
  2 (5, 1005, 'UPI', 'Failed', DATE '2026-09-10', 450.00);
```

1 row created.

```

SQL> INSERT INTO Payment VALUES
  2 (6, 1006, 'Credit Card', 'Successful', DATE '2026-09-12', 1299.00);

1 row created.

SQL> INSERT INTO Payment VALUES
  2 (7, 1007, 'Net Banking', 'Failed', DATE '2026-09-15', 1999.00);

1 row created.

SQL> INSERT INTO Payment VALUES
  2 (8, 1008, 'Debit Card', 'Successful', DATE '2026-09-18', 1049.00);

1 row created.

```

4.Display Payment Records :

```
SQL> SELECT * FROM Payment;
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
1	1001	Credit Card	Successful
01-SEP-26	2098		
2	1002	UPI	Successful
03-SEP-26	1499		
3	1003	Debit Card	Successful
05-SEP-26	2598		

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
4	1004	Cash on Delivery	Successful
07-SEP-26	1999		
5	1005	UPI	Failed
10-SEP-26	450		
6	1006	Credit Card	Successful
12-SEP-26	1299		

```

PAYMENT_ID  ORDER_ID  PAYMENT_METHOD  PAYMENT_STATUS
-----
PAYMENT_D   AMOUNT
-----
          7      1007 Net Banking      Failed
15-SEP-26    1999

          8      1008 Debit Card      Successful
18-SEP-26    1049

8 rows selected.

```

5. Successful Transactions :

```

SQL> SELECT * FROM Payment
  2  WHERE Payment_Status = 'Successful';

```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
1	1001	Credit Card	Successful
01-SEP-26	2098		
2	1002	UPI	Successful
03-SEP-26	1499		
3	1003	Debit Card	Successful
05-SEP-26	2598		
4	1004	Cash on Delivery	Successful
07-SEP-26	1999		
6	1006	Credit Card	Successful
12-SEP-26	1299		
8	1008	Debit Card	Successful
18-SEP-26	1049		

```

6 rows selected.

```

6. Failed Transactions :

```
SQL> SELECT * FROM Payment
      2 WHERE Payment_Status = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
5	1005	UPI	Failed
10-SEP-26	450		
7	1007	Net Banking	Failed
15-SEP-26	1999		

7. Modify Payment Status :

```
SQL> UPDATE Payment
      2 SET Payment_Status = 'Successful'
      3 WHERE Payment_ID = 5;
```

1 row updated.

```
SQL> UPDATE Payment
      2 SET Payment_Status = 'Successful'
      3 WHERE Payment_ID = 7;
```

1 row updated.

8.ANALYZE PAYMENT METHODS :

8.1 COUNT TRANSACTIONS BY PAYMENT METHOD :

```
SQL> SELECT Payment_Method,  
2         COUNT(*) AS Total_Transactions  
3   FROM Payment  
4  GROUP BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_TRANSACTIONS
Credit Card	2
UPI	2
Debit Card	2
Cash on Delivery	1
Net Banking	1

8.2 TOTALAMOUNT BY PAYMENT METHOD :

```
SQL> SELECT Payment_Method,  
2         SUM(Amount) AS Total_Amount  
3   FROM Payment  
4  GROUP BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_AMOUNT
Credit Card	3397
UPI	1949
Debit Card	3647
Cash on Delivery	1999
Net Banking	1999

9.PAYMENT TRANSACTION REPORT :

```
SQL> SELECT
  2     Payment_ID,
  3     Order_ID,
  4     Payment_Method,
  5     Payment_Status,
  6     Payment_Date,
  7     Amount
  8 FROM Payment
  9 ORDER BY Payment_Date;
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
1	1001	Credit Card	Successful
01-SEP-26	2098		
2	1002	UPI	Successful
03-SEP-26	1499		
3	1003	Debit Card	Successful
05-SEP-26	2598		

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
4	1004	Cash on Delivery	Successful
07-SEP-26	1999		
5	1005	UPI	Successful
10-SEP-26	450		
6	1006	Credit Card	Successful
12-SEP-26	1299		

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS
7	1007	Net Banking	Successful
15-SEP-26	1999		
8	1008	Debit Card	Successful
18-SEP-26	1049		

8 rows selected.